

► **THEOREM 1** Squaring Operation on Equations

If both sides of an equation are squared, then the solution set of the original equation is a subset of the solution set of the new equation.

Equation	Solution Set
$x = 3$	$\{3\}$
$x^2 = 9$	$\{-3, 3\}$

This theorem provides us with a method of solving some equations involving radicals. It is important to remember that any new equation obtained by raising both sides of an equation to the same power may have solutions, called **extraneous solutions**, that are not solutions of the original equation. Fortunately though, any solution of the original equation must be among those of the new equation.

When raising both sides of an equation to a power, checking solutions is not just a good idea—it is essential to identify any extraneous solutions.

►► EXPLORE-DISCUSS 1

Squaring both sides of the equations $x = \sqrt{x}$ and $x = -\sqrt{x}$ produces the new equation $x^2 = x$. Find the solutions to the new equation and then check for extraneous solutions in each of the original equations.

EXAMPLE

1

Solving Equations Involving Radicals

Solve:

(A) $x + \sqrt{x - 4} = 4$ (B) $\sqrt{2x + 3} - \sqrt{x - 2} = 2$

SOLUTIONS

(A) $x + \sqrt{x - 4} = 4$

Isolate radical on one side.

$$\sqrt{x - 4} = 4 - x$$

Square both sides.

$$(\sqrt{x - 4})^2 = (4 - x)^2$$

See the upcoming caution on squaring the right side.

$$x - 4 = 16 - 8x + x^2$$

Write in standard form.

$$x^2 - 9x + 20 = 0$$

Factor left side.

$$(x - 5)(x - 4) = 0$$

Use the zero product property.

$$x - 5 = 0 \quad \text{or} \quad x - 4 = 0$$

$$x = 5 \quad \text{or} \quad x = 4$$

CHECK

$x = 5$	$x = 4$
$x + \sqrt{x - 4} = 4$	$x + \sqrt{x - 4} = 4$
$5 + \sqrt{5 - 4} \stackrel{?}{=} 4$	$4 + \sqrt{4 - 4} \stackrel{?}{=} 4$
$6 \neq 4$	$4 \checkmark = 4$

This shows that 4 is a solution to the original equation and 5 is extraneous. The only solution is $x = 4$.